Evaluation Pipeline

Part 1 — Perplexity (PPL)

1. Rebuild model from config; load `best.pt` (`strict=True`)
2. Set `model.eval()` + `torch.no_grad()`
3. Forward pass over split; accumulate CE loss
4. $\mathcal{L} = \frac{1}{N} \sum_i \mathrm{CE}(\hat{y}_i, y_i)$
5. $\mathrm{PPL} = e^{\mathcal{L}}$

lower = better

Core Code (evaluate.py)

```
model.load_state_dict(ckpt["model_state"], strict=True)
results = evaluate(model, dl, criterion, device)
results['layerwise'] = _evaluate_layerwise(model, dl, device)
```

Part 2 — Layerwise Probing (MAE)

Goal: Quantify how much absolute position is linearly decodable from hidden states at each of 10 layers (Emb + 8 Blocks + final LN).

Probing Algorithm

1. Call model with `return_hidden_states=True` $\rightarrow$ yields $\mathbf{H}_l \in \mathbb{R}^{B \times T \times d}$
2. Flatten: $\mathbf{X} = \mathbf{H}_l.\mathrm{reshape}(-1, d)$
3. Targets: $\mathbf{y} = [0, 1, \ldots, T-1]$ repeated per batch
4. Augment: $\tilde{\mathbf{X}} = [\mathbf{X} \mid \mathbf{1}]$
5. Solve **exactly** via `torch.linalg.lstsq` $\rightarrow$ $\mathbf{w}$
6. $\mathrm{MAE}_l = \frac{1}{NT} \|\tilde{\mathbf{X}}\mathbf{w} - \mathbf{y}\|_1$

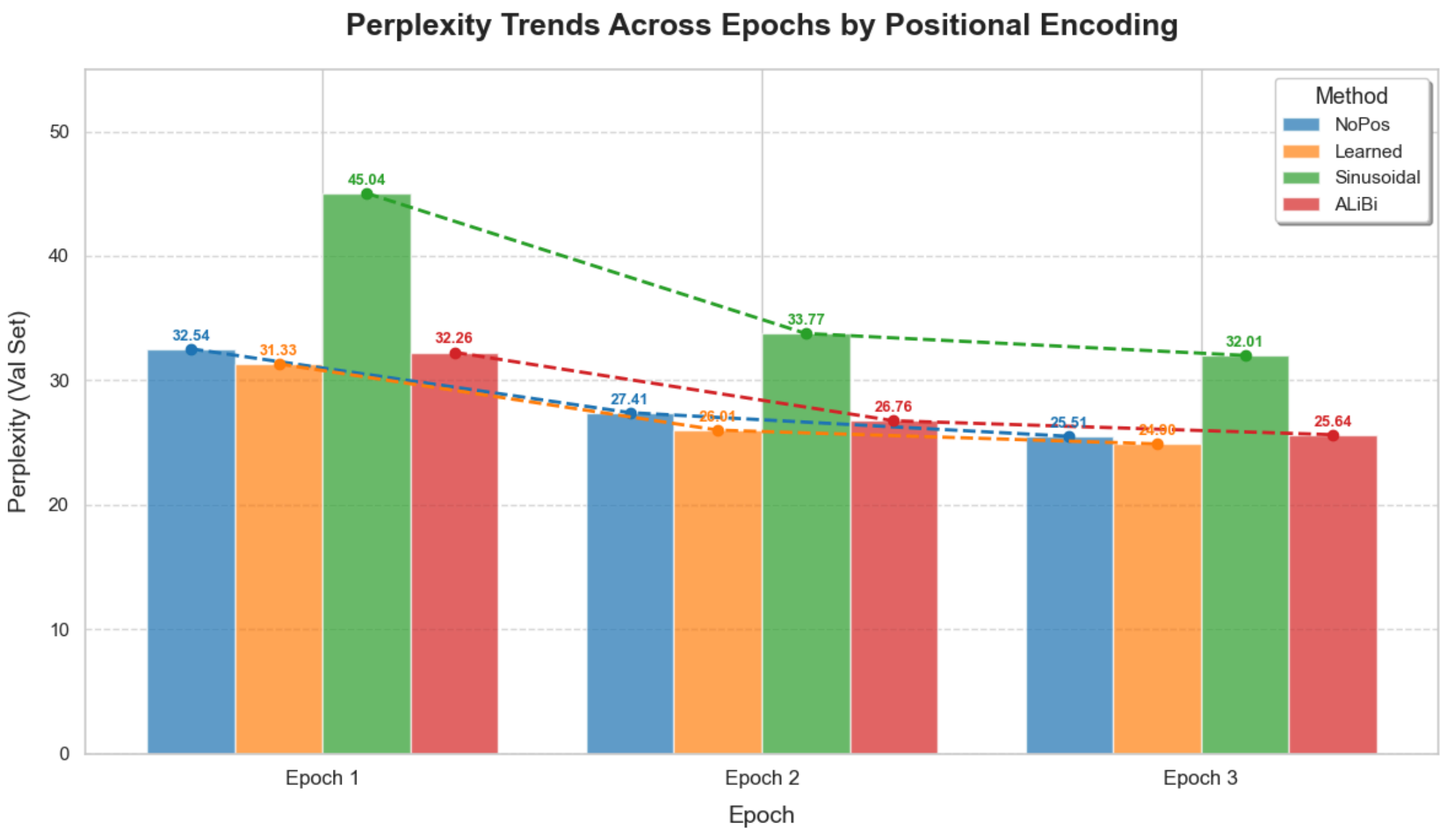
Lower MAE at layer $l \Rightarrow$ stronger positional signal.
Linear probe chosen deliberately: low capacity ensures any decodability is a genuine property of the representation.

Results I — Perplexity Comparison (WikiText-103)

Validation Perplexity

Variant	PPL	vs ALiBi
NoPos	25.51	−0.13
Learned	24.90	−0.73
Sinusoidal	32.01	+6.37
ALiBi	**25.64**	—

Paper Table 1 — 247M params, WikiText-103, 512-token seqs.



Our Replication Setup

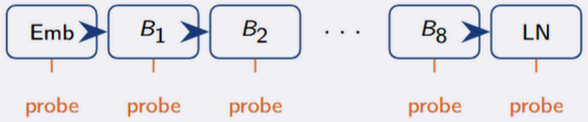
Layers / heads	8 / 8
Sequence len	512
Hardware	Kaggle T4
Epochs	3
Optimizer	AdamW, 3×10^{-4}

Key Takeaway

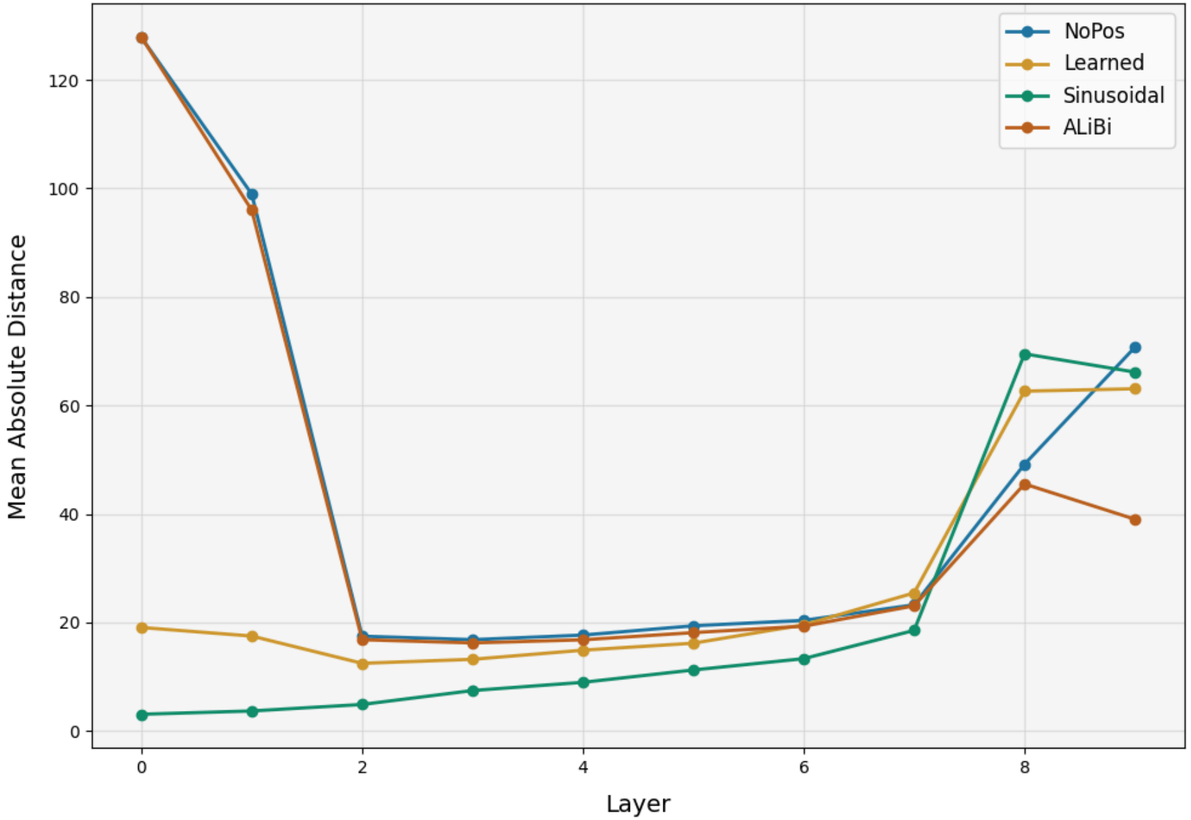
NoPos trails ALiBi by only **-0.13 PPL** — within the ≤ 0.9 random-seed variance of the paper. **No positional encoding is needed to compete.**

Results II — Layerwise Probing & Word-Order Ablation

10 Probe Points (Emb + 8 Blocks + LN)



Stack hidden vecs $\mathbf{X} \in \mathbb{R}^{NT \times d}$; targets $\mathbf{y} = [0 \ldots T-1]$; augment $\tilde{\mathbf{X}} = [\mathbf{X} \mid \mathbf{1}]$; solve via `lstsq`; report $\mathrm{MAE}_l = \frac{1}{NT} \|\hat{\mathbf{y}} - \mathbf{y}\|_1$.



Why Layerwise Probing?

- **NoPos** has no explicit signal — probing tests *implicit* learning
- Reveals *where* in the network positional information emerges
- Linear probe = minimal model ⇒ tests **representation quality**, not probe capacity
- Uses closed-form `torch.linalg.lstsq` (efficient, exact)

Interpretation

Layer 0: NoPos MAE ≈ random — no position in raw embeddings, as expected.

Layers 1–4: MAE collapses: the causal mask lets each token count its predecessors, encoding absolute position.

Mid-layers: NoPos ≈ Learned — implicit encoding matches explicit embeddings.

Final layer: All models shed positional info; representations become semantic (Voita et al. 2019).